

Introduction to Data Analytics

Code: CSIT366

Module 1

Dr. Rashmi Vashisth, Dr. Sakshi Arora, Dr. Ajay Rastogi

Course Objectives:

- Introduce students to rapidly growing field and equip them with some of its basic principles and tools as well as its general mindset.
- To learn concepts, techniques and tools they need to deal with various facets of data science practice, including data collection and integration, exploratory data analysis, predictive modeling, descriptive modeling, data product creation, evaluation, and effective communication.
- Application to solving problems.

Learning Outcomes

- Describe what Data Science is and the skill sets needed to be a data scientist.
- Describe the Data Science Process and how its components interact.
- Differentiation of semantic and discourse in terms of NLP.
- Apply basic machine learning algorithms (Linear Regression, k-Nearest Neighbors
- (k-NN), k-means, Naive Bayes) for predictive modeling. Create effective visualization of given data (to communicate or persuade).

Data All Around

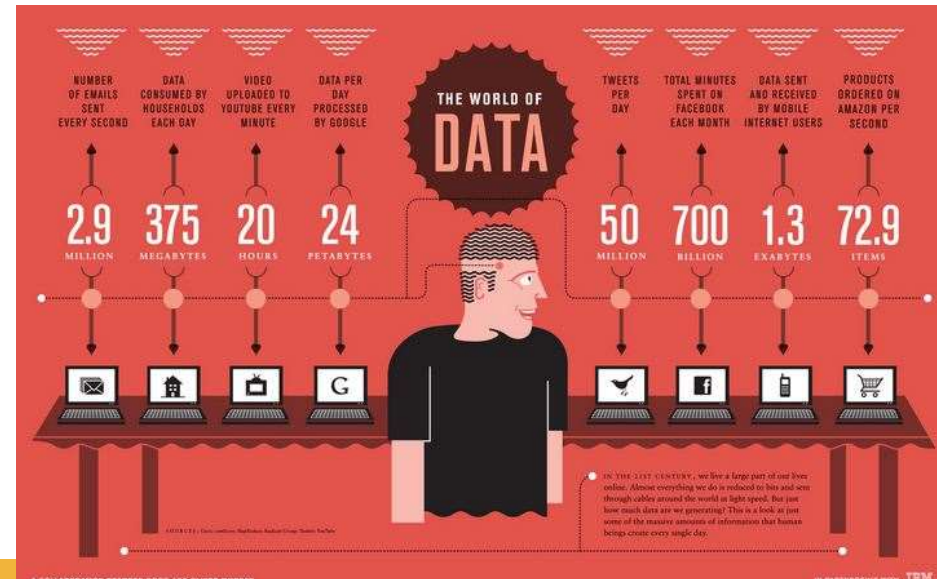
- Lots of data is being collected and warehoused
 - Web data, e-commerce
 - Financial transactions, bank/credit transactions
 - Online trading and purchasing
 - Social Network



How Much Data Do We have?

- Google processes 20 PB a day (2008)
- Facebook has 60 TB of daily logs
- eBay has 6.5 PB of user data + 50 TB/day (5/2009)
- 1000 genomes project: 200 TB

- Cost of 1 TB of disk: \$35
- Time to read 1 TB disk: 3 hrs
(100 MB/s)



What is Big Data?

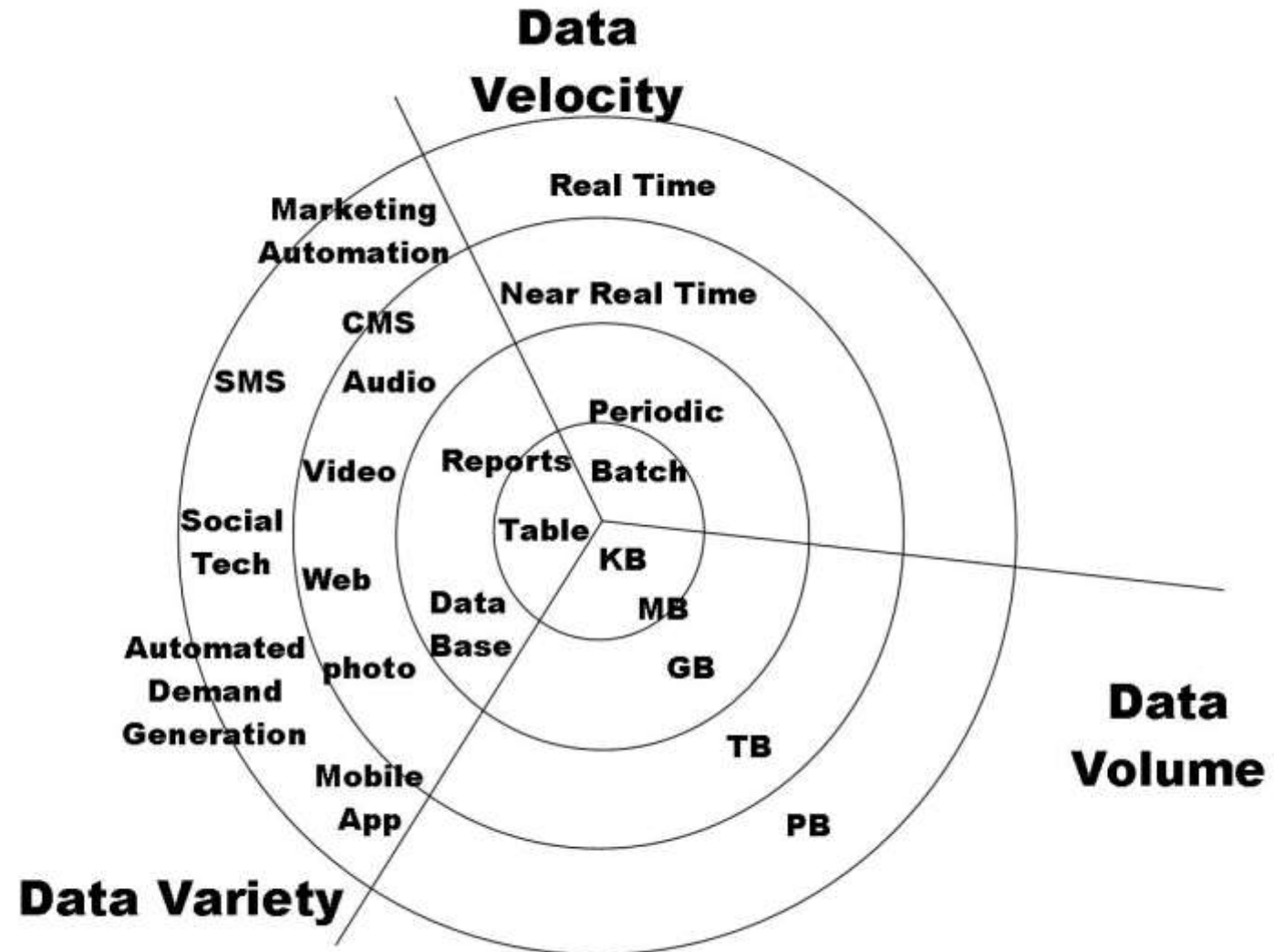
Big Data refers to datasets that are **too large, fast, or complex** to be processed using traditional database systems.

It is commonly characterized by the **5 Vs**:

- **Volume** – Huge amounts of data
- **Velocity** – Rapid generation of data
- **Variety** – Different types of data (structured, semi-structured, unstructured)
- **Veracity** – Data quality and reliability
- **Value** – Useful information obtained from the data

Examples include **social media posts**, **sensor data**, **online transactions**, and IoT data. Technologies such as **Hadoop**, **Spark**, and **NoSQL databases** are commonly used to manage Big Data.

Big Data



Types of Data We Have

- Relational Data (Tables/Transaction/Legacy Data)
- Text Data (Web)
- Semi-structured Data (XML)
- Graph Data
- Social Network, Semantic Web (RDF), ...
- Streaming Data
- You can afford to scan the data once

What To Do With These Data?

- Aggregation and Statistics
 - Data warehousing and OLAP
- Indexing, Searching, and Querying
 - Keyword based search
 - Pattern matching (XML/RDF)
- Knowledge discovery
 - Data Mining
 - Statistical Modeling

Big Data and Data Science

- “... the great job in the next 10 years will be statisticians,” Hal Varian, Google Chief Economist
- The U.S. will need 140,000-190,000 predictive analysts and 1.5 million managers/analysts by 2018. McKinsey Global Institute's June 2011
- New Data Science institutes being created or repurposed – NYU, Columbia, Washington, UCB,...
- New degree programs, courses, boot-camps:
 - e.g., at Berkeley: Stats, I-School, CS, Astronomy...
 - One proposal (elsewhere) for an MS in “Big Data Science”

What is Data Science?

- Data Science is an **interdisciplinary field** that uses **statistics, machine learning, programming, and domain knowledge** to analyze data and solve real-world problems.
- Data science is applicable to both – big and small data

The typical Data Science workflow includes:

1. Data collection
2. Data cleaning
3. Data analysis
4. Model building (Machine Learning)
5. Visualization
6. Decision-making

Common tools include: **Python, R, Pandas, NumPy, Scikit-learn, TensorFlow, and Matplotlib.**

Applications of Data Science?

- Big data helps in decision makers for many industries such as science, engineering, economics, politics, finance, and education.
- **Computer Science**
 - Pattern recognition, visualization, data warehousing, High performance computing, Databases, AI
- **Mathematics**
 - Mathematical Modeling
- **Statistics**
 - Statistical and Stochastic modeling, Probability.

expectations



Plateau will be reached in:

○ less than 2 years

● 2 to 5 years

● 5 to 10 years

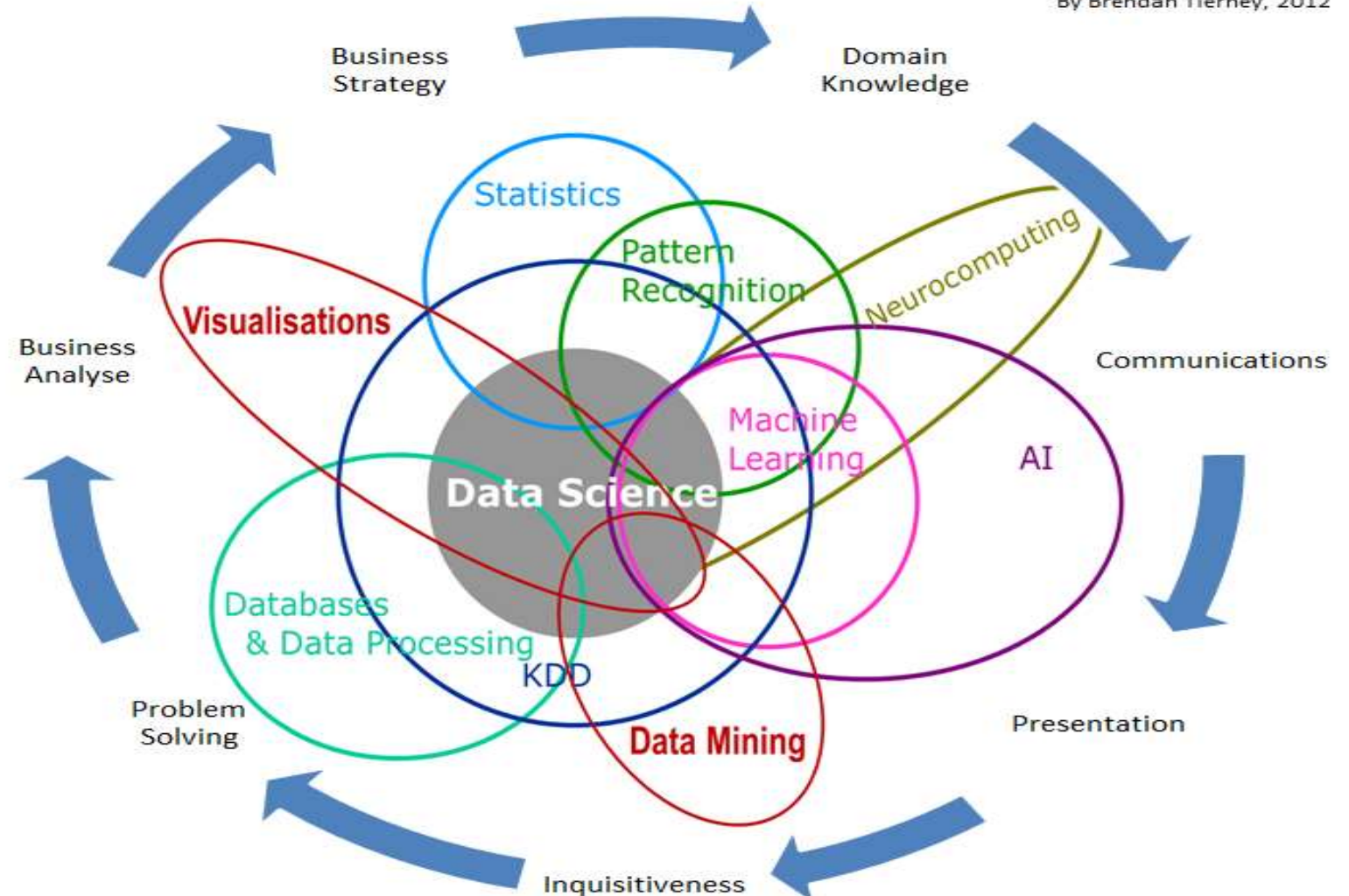
▲ more than 10 years

⊗ obsolete before plateau

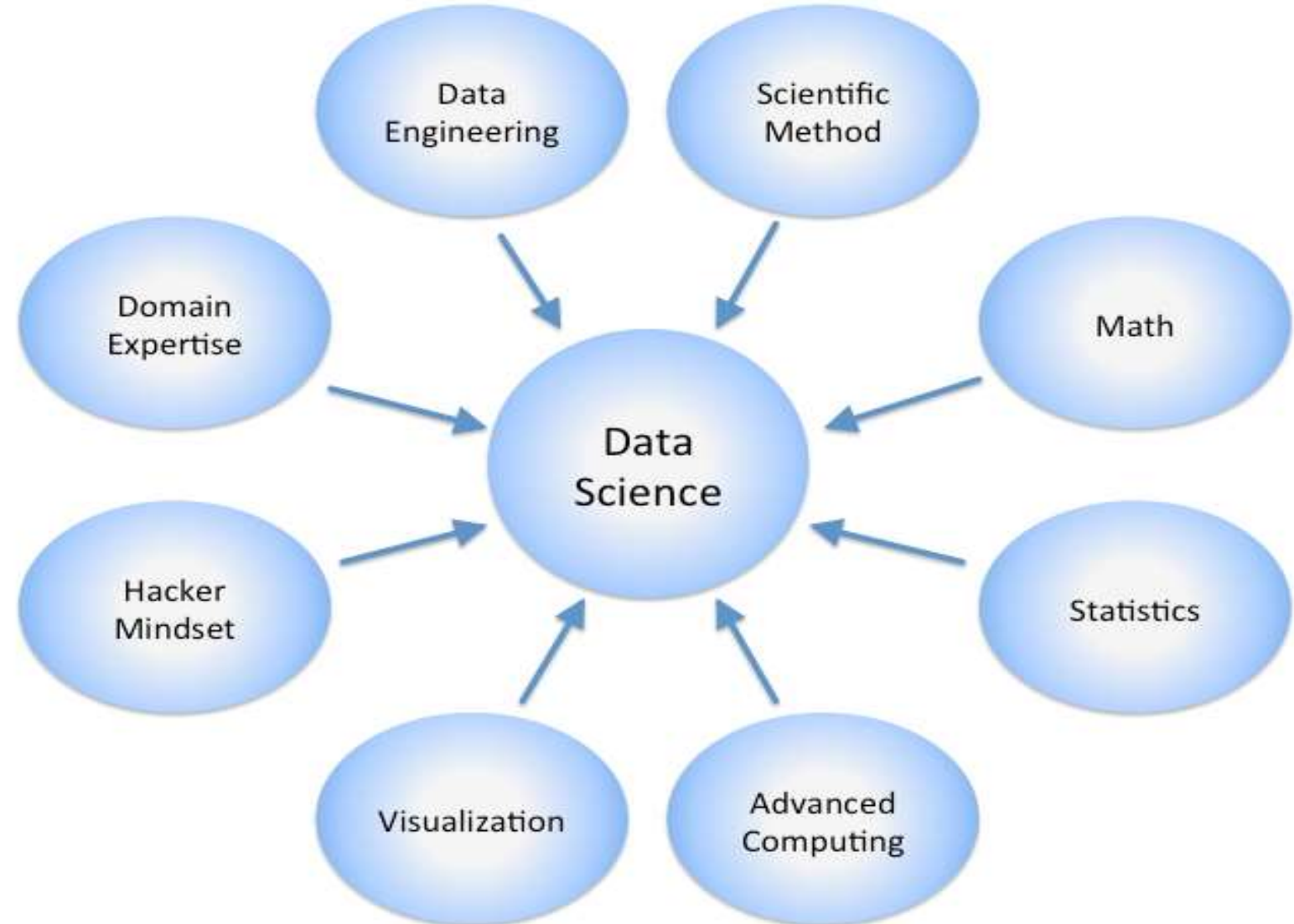
Data Science Is Multidisciplinary

By Brendan Tierney, 2012

Data Science



Data Science



Real Life Examples

- Companies learn your secrets, shopping patterns, and preferences
- Data Science and election (2008, 2012)
 - 1 million people installed the Obama Facebook app that gave access to info on “friends”

Relationship Between Big Data and Data Science

- Big Data provides the **raw data**, while Data Science provides the **methods and algorithms** to analyze that data.

For example:

- An e-commerce company collects **terabytes of customer purchase data** (Big Data).
- A Data Scientist analyzes this data to recommend products, predict customer behavior, and detect fraud (Data Science).

Thus, **Big Data is the fuel, and Data Science is the engine that converts that fuel into valuable insights.**

Data Scientists

- Data Scientist
 - The great Job of the 21st Century
- They find stories, extract knowledge. They are not reporters



Data Scientists

- Data scientists are the key to realizing the opportunities presented by big data. They bring structure to it, find compelling patterns in it, and advise executives on the implications for products, processes, and decisions



What do Data Scientists do?

- National Security
- Cyber Security
- Business Analytics
- Engineering
- Healthcare
- And more

Concentration in Data Science

- Mathematics and Applied Mathematics
- Applied Statistics/Data Analysis
- Solid Programming Skills (R, Python, Julia, SQL)
- Data Mining
- Data Base Storage and Management
- Machine Learning and discovery